COURSE OUTLINE

Faculty of Computing and Informatics · Sample university for testing Lyceum (fictional)

Course code	CSC2244
Course title	Data Analytics and Visualisation
Programme	Bachelor of Computer Science (Hons)
Credit value	3
Year / Semester	Year 2, Semester 1
Academic session	2026/2027 Semester 1
Teaching weeks	14
Prerequisite	CSC1104 Programming Fundamentals; MAT1203 Statistics
Total student learning time (SLT)	120 hours: 42 contact hours (28 lecture, 14 lab), 78 independent learning

Synopsis

This course introduces the process of turning raw data into decisions: cleaning and preparing data, exploratory analysis, descriptive statistics, building dashboards and communicating findings to non-technical stakeholders. Students work with Python (pandas, matplotlib) and a business intelligence tool on real public datasets.

Course Learning Outcomes (CLO)

At the end of this course, students should be able to:

CLO	Statement	PLO
CLO1	Explain the stages of the data analytics life cycle and the role of each stage in a business decision.	PLO1
CLO2	Apply data cleaning and transformation techniques to prepare a raw dataset for analysis using Python.	PLO2
CLO3	Analyse a dataset with descriptive statistics and exploratory visualisation to identify patterns and outliers.	PLO2
CLO4	Design an interactive dashboard that answers a stated business question for a non-technical audience.	PLO3
CLO5	Evaluate the reliability and ethical risks of a data-driven recommendation, including privacy and bias.	PLO4

Programme Learning Outcomes (PLO) addressed

PLO1 Knowledge: apply knowledge of computing fundamentals to solve problems.
PLO2 Practical skills: use modern tools and techniques to analyse and solve computing problems.
PLO3 Communication: communicate technical information effectively to a range of audiences.
PLO4 Ethics and professionalism: apply ethical principles and professional responsibilities in computing practice.

Weekly Topics

Week	Topic	Contact hours	CLO
1	Introduction to data analytics and the analytics life cycle	3	CLO1
2	Data sources, formats and collecting data	3	CLO1

3	Python for data analysis: pandas basics	3	CLO2
4	Data cleaning: missing values, duplicates and types	3	CLO2
5	Data transformation, joins and reshaping	3	CLO2
6	Descriptive statistics and distributions	3	CLO3
7	Exploratory data analysis and outlier detection	3	CLO3
8	Mid-semester break	0	-
9	Principles of visual design and chart selection	3	CLO3, CLO4
10	Building dashboards in a business intelligence tool	3	CLO4
11	Storytelling with data for stakeholders	3	CLO4
12	Data privacy, bias and ethics in analytics	3	CLO5
13	Group project clinic	3	CLO4, CLO5
14	Project presentations and course review	3	CLO4, CLO5

Assessment

Component	Weight	CLO
Lab exercises (4)	20%	CLO2, CLO3
Quiz (2)	10%	CLO1, CLO3
Group dashboard project and presentation	30%	CLO4, CLO5
Final examination	40%	CLO1, CLO3, CLO5

Career Prospects

Graduates of this course are prepared for roles such as data analyst, business intelligence analyst and reporting analyst in Malaysia and the region.

Main References

McKinney, W. (2022). Python for Data Analysis, 3rd ed. O'Reilly.
Knaflic, C. N. (2015). Storytelling with Data. Wiley.

A fictional outline written to test Lyceum's outline import. Not an official document of any university.